# Arista Access Node BAS-ACC-78 Basavanagudi POP Optical Transport Failure Resolution

**Node:** BAS-ACC-78 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_6966

## Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Arista Access Node BAS-ACC-78 at Basavanagudi POP is caused by the degradation of optical signals transmitted through the Optical Transport Network (OTN) infrastructure. This failure domain occurs at the Access Node level due to the role's responsibility in aggregating and transmitting customer traffic to the core network. The root cause of this failure can be attributed to the physical degradation of SFP/XFP transceivers, fiber cuts or breaks, or power loss in the optical signal. These issues can lead to a cascade of failures in the control plane, resulting in the ALARM_6966 alarm code being raised on the node. It is essential to diagnose and resolve these issues promptly to ensure continuous network availability and prevent potential data losses.

## Pre-requisite Checks

- Prerequisite: Verify the status of redundant links and ensure that they are operational before proceeding with the resolution steps.

- Prerequisite: Ensure that remote hands are dispatched to the Basavanagudi POP site to assist with physical checks and repairs, if necessary.

## Potential Root Causes

| Cause | Probability |
|---|---|
| SFP/XFP Transceiver Degradation | High |
| Fiber Cut/Break | Medium |
| Rx/Tx Power Loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface information, including link status and optical signal quality, Description: This command displays the interface status, including link status and optical signal quality, to help identify potential issues with the SFP/XFP transceivers or fiber connections.

Command: show optics, Expected Output: Optics information, including temperature, voltage, and optical signal quality, Description: This command displays optics information, including temperature, voltage, and optical signal quality, to help diagnose issues with the SFP/XFP transceivers.

Command: show log | include ALARM_6966, Expected Output: Log messages related to the ALARM_6966 alarm code, Description: This command
```

```
displays log messages related to the ALARM_6966 alarm code, providing
valuable information on the timing and context of the failure.
```

## Resolution Steps

Step 1: Step: Replace SFP/XFP Transceiver (if faulty), Command: replace interface with SFP-10G-SR, Description: Replace the faulty SFP/XFP transceiver with a new one to restore the optical signal quality and prevent further failures.

Step 2: Step: Repair Fiber Cut/Break (if applicable), Command: no shutdown, Description: Repair the fiber cut or break to restore the optical signal quality and prevent further failures.

Step 3: Step: Adjust Rx/Tx Power Settings (if applicable), Command: set optics power-adjust, Description: Adjust the Rx/Tx power settings to optimize the optical signal quality and prevent further failures.

## Post-Check Verification

```
Command: show interface status, Expected Output: Interface
information, including link status and optical signal quality,
Description: Verify that the interface status and optical signal
quality have been restored to normal after the resolution steps.

Command: show optics, Expected Output: Optics information, including
temperature, voltage, and optical signal quality, Description: Verify
that the optics information, including temperature, voltage, and
optical signal quality, has been restored to normal after the
resolution steps.

Command: show log | include ALARM_6966, Expected Output: Log messages
related to the ALARM_6966 alarm code, Description: Verify that the
ALARM_6966 alarm code is no longer being raised after the resolution
steps.
```

## Roll-back Procedures

- Step: Restore original SFP/XFP Transceiver (if replaced), Command: replace interface with original SFP-10G-SR, Description: Restore the original SFP/XFP transceiver to its original state in case the replacement SFP/XFP transceiver did not resolve the issue.

- Step: Re-apply previous fiber repairs (if applicable), Command: shutdown, Description: Re-apply the previous fiber repairs in case the repairs did not resolve the issue.

- Step: Revert Rx/Tx Power Settings (if adjusted), Command: clear optics power-adjust, Description: Revert the Rx/Tx power settings to their original state in case the adjustments did not resolve the issue.